

## Sonar qube Backend .Net

Step1: Add .sonarqube folder in the project [right now this folder is in Clinic Visit Appointment booking System Backend-User Microservice, Schedule Microservice]

Step2: Open command prompt and go till Test project and install this packages

⇒ dotnet add package coverlet.msbuild

⇒ dotnet add package FluentAssertions

Step3: Run StartSonar.bat in Path:C:\New folder (2)\sonarqube-9.2.4.50792\sonarqube-9.2.4.50792\bin\windows-x86-64

Open localhost:9000 in browser-SonarQube login:admin password:admin or admin123

And create a project manually give name and generate key and create a bat file using that generated key in next step

The screenshot displays the SonarQube web interface at localhost:9000/projects. The interface includes a top navigation bar with tabs for Projects, Issues, Rules, Quality Profiles, Quality Gates, and Administration. A search bar is present on the right. The main content area shows a list of projects. Two projects are visible:

- ScheduleMicroserviceBackend**: Status: Passed, Last analysis: 18 minutes ago. Metrics: Bugs (A), Vulnerabilities (A), Hotspots Reviewed (A), Code Smells (A), Coverage (100%), Duplications (0.0%), Lines (588 XS C#).
- UserMicroserviceBackend**: Status: Passed, Last analysis: 25 minutes ago. Metrics: Bugs (A), Vulnerabilities (A), Hotspots Reviewed (A), Code Smells (A), Coverage (100%), Duplications (0.0%), Lines (652 XS C#).

A yellow warning banner at the bottom states: "Embedded database should be used for evaluation purposes only. The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine."

be. Be.Cognizant - Home x How do you want to create your x .NET Core Code Quality with Cov x msbuild - How to make Sonarq x +

localhost:9000/projects/create?mode=manual

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration ? Search for projects... A

## Create a project

All fields marked with \* are required

**Project display name \***

ClinicVisitBackend ✓

Up to 255 characters. Some scanners might override the value you provide.

**Project key \***

ClinicVisitBackend ✓

The project key is a unique identifier for your project. It may contain up to 400 characters. Allowed characters are alphanumeric, '-' (dash), '\_' (underscore), '.' (period) and ':' (colon), with at least one non-digit.

**Set Up**

**Embedded database should be used for evaluation purposes only**

The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.

SonarQube™ technology is powered by SonarSource SA  
Community Edition - Version 9.2.4 (build 50792) - LGPL v3 - Community - Documentation - Plugins - Web API

## Select Locally

be. Be.Cognizant - Home x ClinicVisitBackend x .NET Core Code Quality with Cov x msbuild - How to make Sonarq x +

localhost:9000/dashboard?id=ClinicVisitBackend

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration ? Search for projects... A

ClinicVisitBackend ☆ master

Overview Issues Security Hotspots Measures Code Activity Project Settings Project Information

Do you want to integrate with your favorite CI? Choose one of the following tutorials.

  
With Jenkins

  
With GitHub Actions

  
With Bitbucket Pipelines

  
With GitLab CI

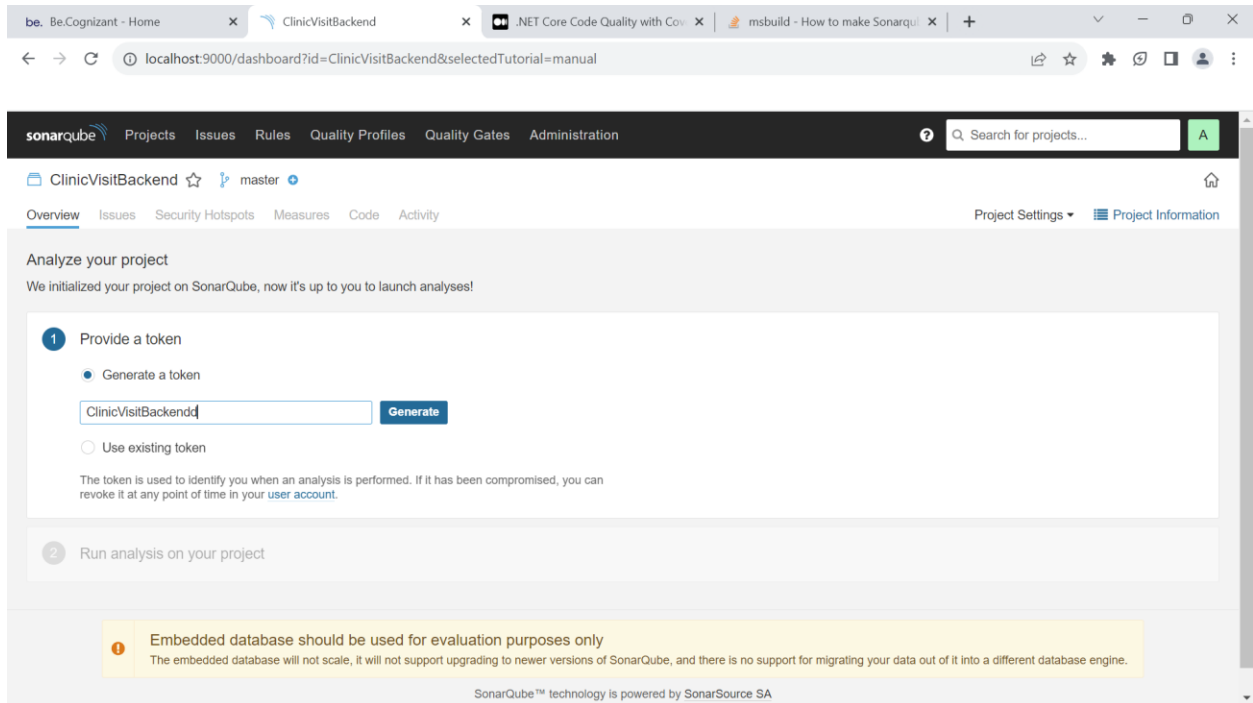
  
With Azure Pipelines

  
Other CI

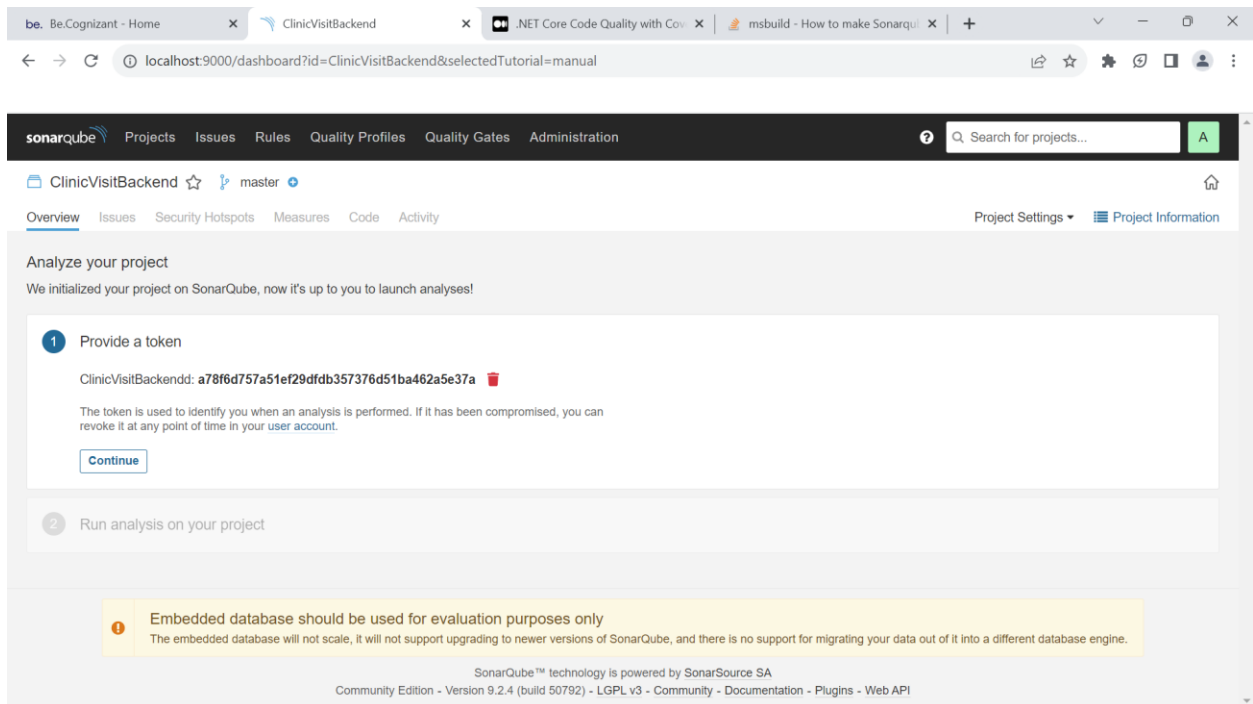
Are you just testing or have an advanced use-case? Analyze your project locally.

  
Locally

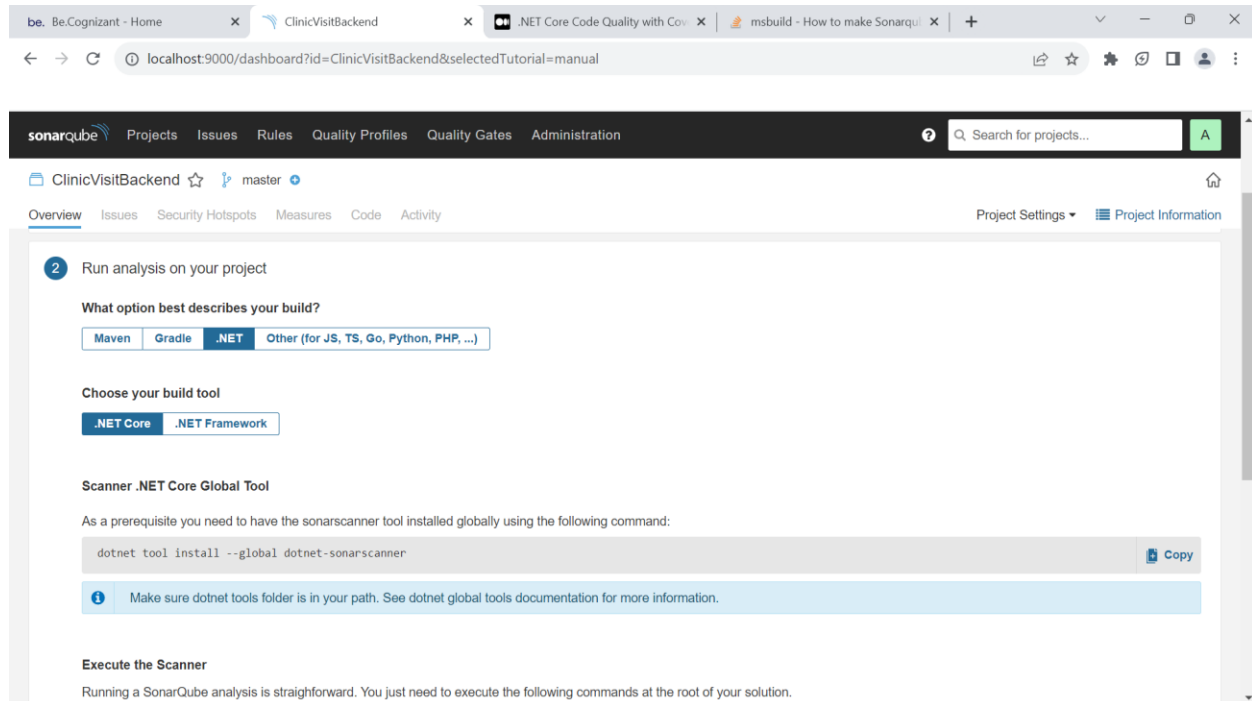
Give Key and generate token by clicking generate



Click continue



Select .Net and open command prompt and install tool that shown in the image if you are doing it for .net first time else no need



Using that token now we will create the sonar scan.bat file

Step4: create a sonarscan.bat file

```
cd C:\Users\921611\source\repos\Clinic Visit Appointment Booking System\UserMicroservice
```

```
dotnet test .\UserMicroServiceTest\UserMicroServiceTest.csproj /p:CollectCoverage=true  
/p:CoverletOutputFormat=opencover
```

```
dotnet sonarscanner begin /k:"UserMicroserviceBackend" /d:sonar.host.url="http://localhost:9000"  
/d:sonar.cs.opencover.reportsPaths=./UserMicroServiceTest/coverage.opencover.xml  
/d:sonar.login="541d6f1cc69c126fa5dc5fca106436424d1c6b9b"
```

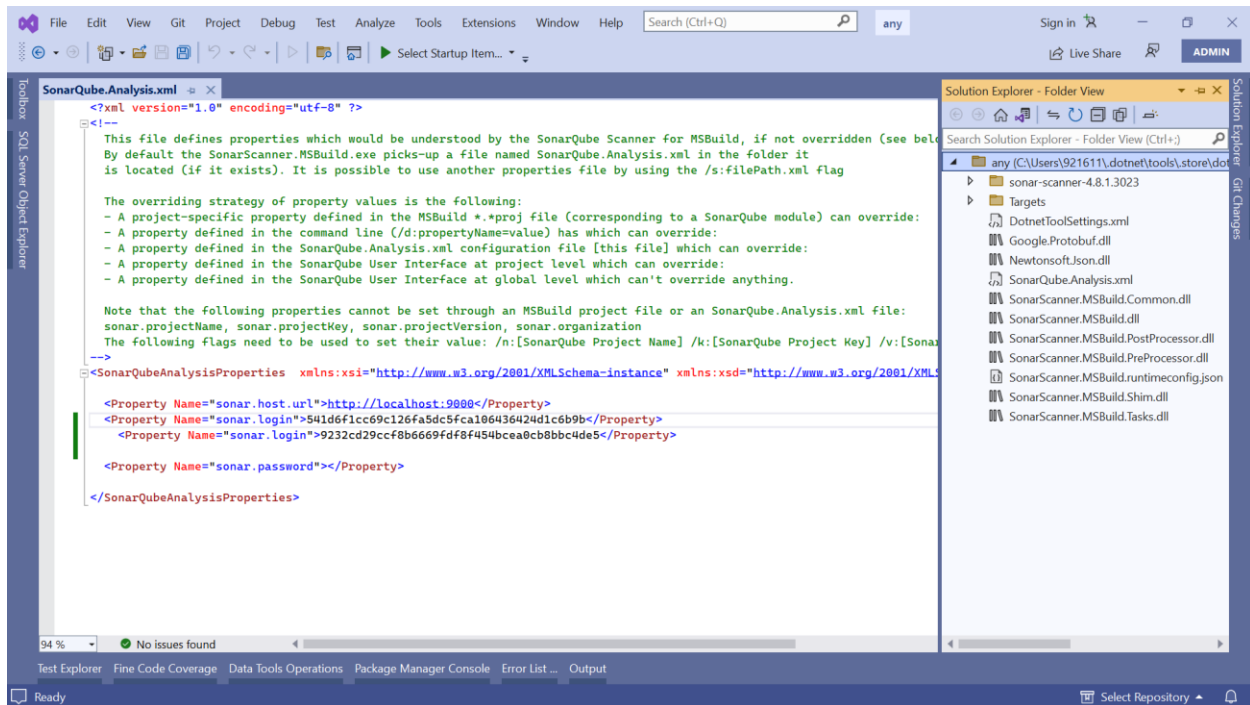
```
dotnet build
```

```
dotnet sonarscanner end /d:sonar.login="541d6f1cc69c126fa5dc5fca106436424d1c6b9b"
```

Step5:

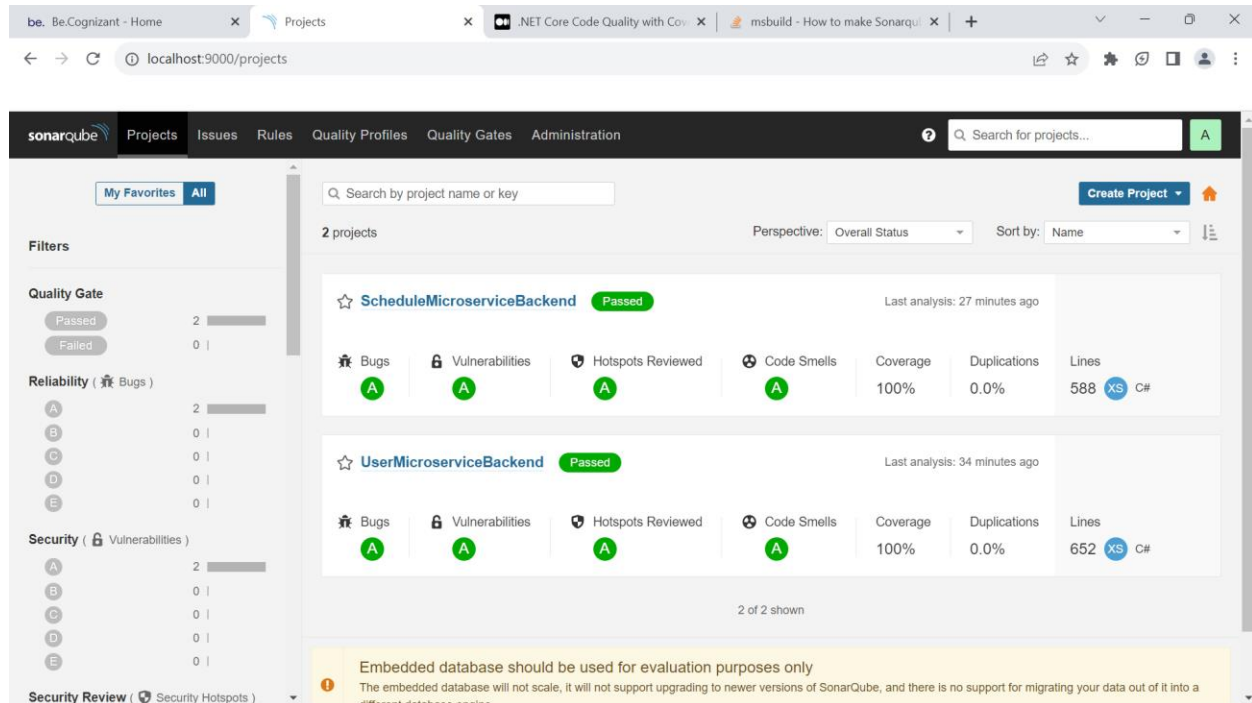
Path `C:\Users\921611\dotnet\tools\.store\dotnet-sonarscanner\5.13.1\dotnet-sonarscanner\5.13.1\tools\net5.0\any`

Open any folder in visual studio and open SonarQube.Analysis.xml file and uncomment this code and add login generated keys of your projects Created in SonarQube(localhost:9000)



Step 5: Now run/ double click on the sonarscan.bat file

And see your projects been added to Sonar qube



Step6: For excluding Some files, a file or entire project from Code Coverage

1. To exclude a project from SonarQube Analysis from csproj we can achieve by adding the below code in .csproj of that project

```
<PropertyGroup>
<!-- Exclude the project from analysis -->
<SonarQubeExclude>true</SonarQubeExclude>
</PropertyGroup>
```

2. To exclude a file from a project we can achieve by adding the below code in .csproj of that project

```
<ItemGroup>
<SonarQubeSetting Include="sonar.coverage.exclusions">
<Value>*/FileName.cs</Value>
</SonarQubeSetting>
</ItemGroup>
```

3. To Exclude multiple files from a project we can achieve by adding the below code in .csproj of that project

```
<ItemGroup>
<SonarQubeSetting Include="sonar.coverage.exclusions">
<Value>*/FileName1.cs, */FileName2.cs</Value>
</SonarQubeSetting>
</ItemGroup>
```

Reference WebSite like

<https://medium.com/tribalscale/net-core-code-quality-with-coverlet-and-sonarqube-part-2-141617f83d6f>